Chapter 1: Similarity

1. In ΔABC, DE ∥ BC. If AD = 3 cm, DB = 6 cm and EC = 8 cm, find AE.

2. In ΔPQR, XY ∥ QR. If PX = 4 cm, XQ = 6 cm and PR = 15 cm, find PY.

3. If ΔABC ~ΔPQR and AB = 6 cm, PQ = 9 cm, BC = 8 cm, find QR.

4. If area(ΔABC) : area(ΔPQR) = 16 : 25, find AB : PQ.

5. A line parallel to one side of a triangle divides other sides in ratio 3 : 5. If one segment is 6 cm, find the other.

6. In ΔABC, PQ ∥ BC. If area(APQ) : area(ABC) = 1 : 4, find AP : AB.

7. If ratio of perimeters of two similar triangles is 5 : 7, find ratio of their areas.

8. Prove that a line drawn parallel to one side of a triangle divides other two sides proportionally.

9. Prove the converse of Basic Proportionality Theorem.

10. In ΔABC, AB = 12 cm, AD = 4 cm, AC = 15 cm, DE ∥ BC. Find AE.